AMERICAN UNIVERSITY OF PHNOM PENH STUDY LOCALLY. LIVE GLOBALLY	American University of Phnom Penh
	School of Digital Technologies
	ITM-390 Machine Learning

Final Project Report

Project Information			
Project title	English	Unsupervised Topic Discovery in Cambodian Reddit Discussion	
	Khmer	ការស្វែងរកប្រធានបទដោយមិនត្រូវការទុកមុនក្នុងការជជែកគ្នានៅលើ Reddit របស់កម្ពុជា	
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I. Introduction

Social media platforms have generated large volumes of user-created content that reflects public opinions, concerns, and emerging trends. However, transforming this unstructured text into actionable insights is challenging due to the scale of data, noisy language, rapidly changing discussions, and uncertainty in data quality. These problems are well documented in social media analytics, especially in the early stages of research such as topic discovery, data collection, and data preparation [1].

Reddit provides a valuable space for topic discovery because discussions are community-driven and organized into subreddits. In Cambodia, online communities increasingly influence public discourse and consumer behavior. Research in Cambodia also shows that social media is a primary channel for discovering information and shaping perceptions, which highlights the importance of understanding what Cambodian users discuss online [8]. Despite this relevance, there is limited academic work that focuses on unsupervised topic discovery in Cambodian-focused Reddit communities.

This project examines an unsupervised machine learning pipeline to uncover latent topics in posts from r/Cambodia. The pipeline uses a sentence transformer model (all-mpnet-base-v2) for semantic embeddings, UMAP for dimensionality reduction, HDBSCAN for density-based clustering, and c-TF-IDF for interpretable topic labeling. The goal is to produce meaningful topic clusters and representative keywords that summarize community discussions and support real-world use cases such as trend monitoring and content moderation [1], [10].

II. Literature review

Social media analytics (SMA) refers to the process of collecting and extracting user opinions from comments, posts, and text, then analyzing the content. Stieglitz et al. [1] argue that major research gaps exist in the early stages of SMA, particularly in topic discovery, data collection, and preparation. Social media data presents challenges including the large scale of datasets, rapid updates, bad quality data, and noise.

Latent Dirichlet Allocation (LDA) and Latent Semantic Analysis (LSA) have been introduced to address topic modeling. LDA is a probabilistic generative model that assumes documents are mixtures of diverse topics [3]. Even though LDA is widely used, it struggles with short documents, especially user comments on social media. LSA was introduced to address this: Deerwester et al. [4] applied Singular Value Decomposition (SVD) to a term-document matrix to reduce dimensionality and capture latent semantic relationships. Landauer et al. [5] further demonstrated that LSA effectively models conceptual similarity between documents. Unlike LDA, LSA is algebraic rather than probabilistic and is computationally simpler for moderate datasets [9]. However, both LSA and LDA rely on bag-of-words representations that ignore word order and contextual nuance.

Devlin et al. [6] introduced BERT, a deep bidirectional transformer model that generates contextualized word representations. Unlike traditional methods, BERT captures semantic

meaning based on surrounding context. Embedding-based topic modeling frameworks such as BERTopic combine BERT embeddings with density-based clustering and keyword extraction, offering improved performance on short and noisy text data [10].

For clustering, K-Means assumes spherical clusters and requires predefining the number of clusters, which may not suit dynamic social media topics. McInnes et al. [7] proposed HDBSCAN, a hierarchical density-based clustering algorithm capable of detecting clusters of varying densities and labeling outliers as noise, making it especially suitable for Reddit discussions where topic distributions may not be uniform.

Research by Em et al. [8] demonstrates that social media plays a significant role in shaping information discovery and consumer behavior in Cambodia. Despite this importance, there is limited comparative research applying and evaluating both classical and transformer-based unsupervised topic discovery methods specifically on Cambodian Reddit data.

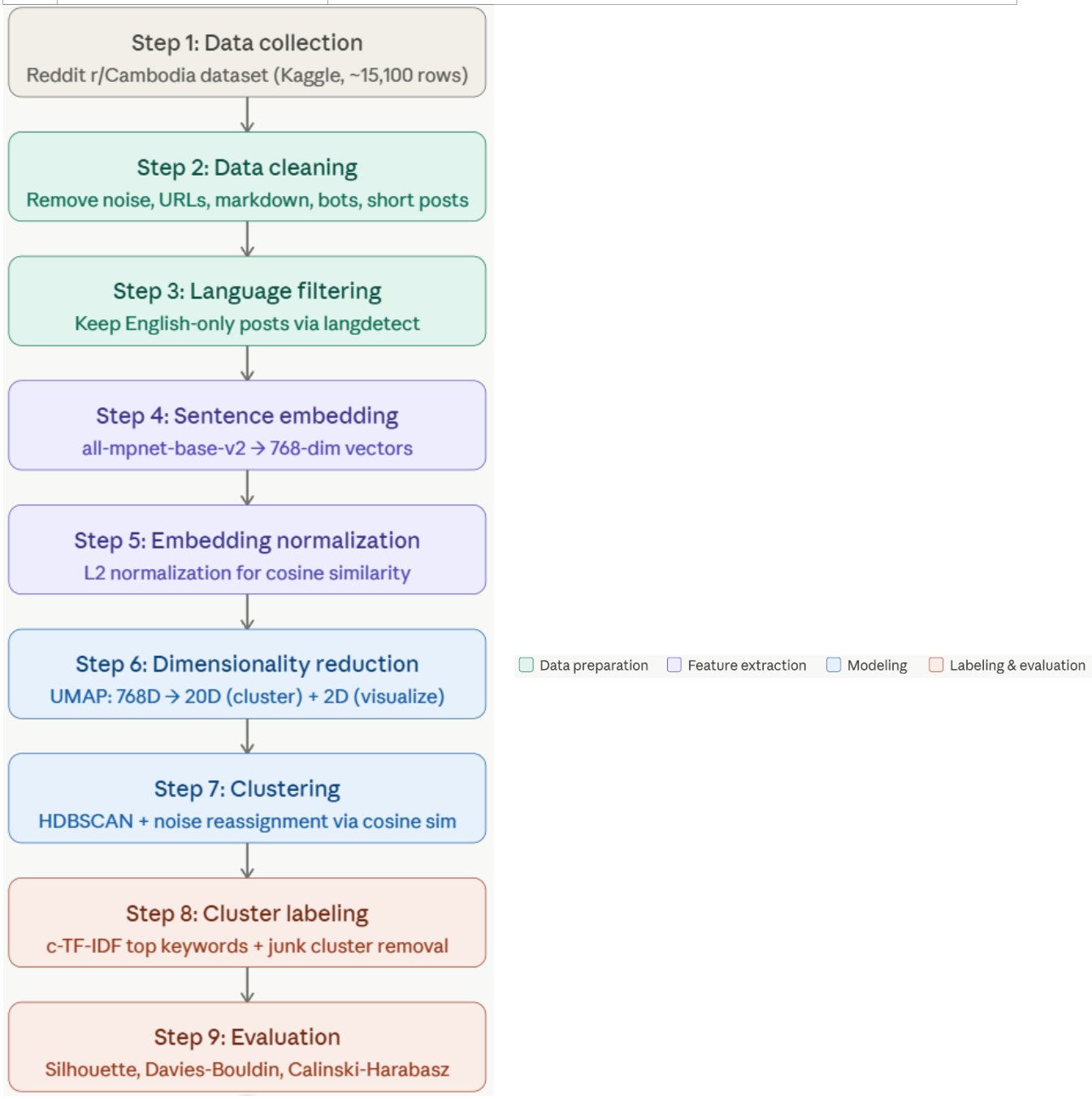
III. Methodology

This study uses a fully unsupervised NLP pipeline to discover latent topics from the r/Cambodia subreddit. Since the data has no labels, no train/test split is performed. All data is used to fit the model. The pipeline consists of seven sequential steps as shown in Table 1 below.

Table 1. Pipeline Steps Overview

#	Step	Details
1	**Data Cleaning**	Remove [removed] entries, URLs, markdown, Reddit artifacts, boilerplate bot/mod text, special characters, contractions expansion, and short tokens (≤2 chars or ≤5 words after cleaning).
2	**Sentence Embedding**	Encode all English comments using all-mpnet-base-v2 (sentence-BERT) into 768-dimensional dense vectors.
3	**Embedding Normalization**	Apply L2 normalization so all vectors have unit length, ensuring consistent cosine similarity comparisons.
4	**Dimensionality Reduction (UMAP)**	Reduce embeddings to 20D (for clustering) and separately to 2D (for visualization), using cosine metric.
5	**Clustering (HDBSCAN)**	Cluster the 20D embeddings. Noise points are reassigned to nearest centroid if cosine similarity > 0.5.

6	**Cluster Labeling (c-TF-IDF)**	Extract top 10 TF-IDF terms per cluster as topic keywords. Flag and remove junk clusters.
7	**Evaluation**	Compute Silhouette Score, Davies-Bouldin Score, and Calinski-Harabasz Score on final clusters.



Dataset

This study utilizes the publicly available dataset from Kaggle titled "Reddit r/Cambodia Subreddit Dataset 2026" (bwandowando/reddit-rcambodia-subreddit-dataset-2026). The dataset consists of 12 columns with approximately 15,100 rows of comments from the r/Cambodia subreddit. The primary column used is body, which contains the raw comment text. Since the data is unlabeled, the entire dataset is used for unsupervised model fitting without any train/test splitting.

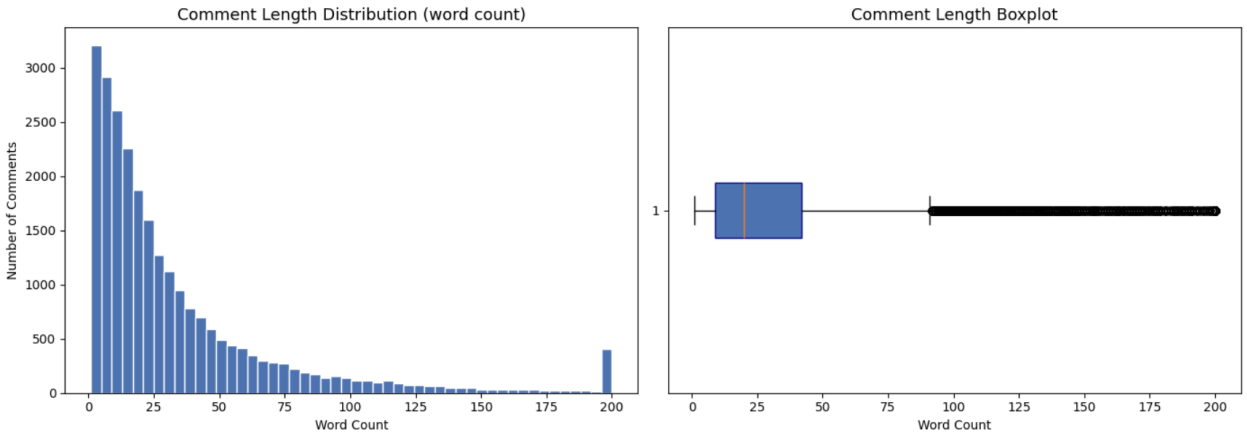
Embedding Model Selection

The all-mpnet-base-v2 model from sentence-transformers was selected as the embedding model. This model produces 768-dimensional sentence embeddings and is optimized for semantic similarity tasks. While the r/Cambodia community contains some Khmer text, language detection revealed that the vast majority of posts are in English. Therefore, filtering to English-only posts and using a high-quality English sentence-BERT model was more appropriate than using a multilingual model such as LaBSE, which would introduce unnecessary complexity for a predominantly English corpus.

IV. Experiments

Exploratory Data Analysis

Before model development, we performed EDA on the raw dataset to understand its structure and quality. The dataset contained 15,100 rows and 12 columns. A significant number of rows had body values marked as "[removed]", which were filtered out as they carry no content. Comment length analysis showed that most posts are short (under 50 words), with a right-skewed distribution.



A small number of very long comments pull the average up. This confirmed the need for a sentence-level embedding model rather than bag-of-words methods. Language detection using langdetect revealed that the corpus is predominantly English, with minor misdetections (e.g., Afrikaans, Slovenian) arising from very short posts. The timestamp column (created_utc) was found to contain no valid data, so post activity over time analysis was not performed. Word frequency analysis after cleaning showed that the most common terms relate to everyday life in

Cambodia is travel, food, housing, and local culture which providing an early qualitative signal of expected topic clusters.

Data Preprocessing Configuration

The cleaning pipeline applied the following steps in order: (1) lowercase conversion, (2) contraction expansion (e.g., "don't" → "do not"), (3) removal of Reddit artifacts (r/, u/, [deleted], [removed]), (4) URL removal, (5) markdown stripping, (6) boilerplate phrase filtering using regex patterns targeting bot and moderator messages, (7) special character removal, (8) whitespace normalization, and (9) removal of tokens that are 2 characters or fewer, contain digits, or appear in the custom stop word list. A final minimum word-length filter discarded any post with fewer than 5 words after cleaning to remove low-information entries.

UMAP Configuration

UMAP was run twice independently. For clustering, n_neighbors=50, n_components=20, min_dist=0.1, and metric="cosine" were used to reduce from 768D to 20D. For visualization, n_neighbors=15, n_components=2, and min_dist=0.1 were used to project to 2D. The two projections were computed separately to avoid the 2D visualization distorting the clustering space.

HDBSCAN Configuration

HDBSCAN was configured with min_cluster_size=50, min_samples=5, cluster_selection_method="leaf", cluster_selection_epsilon=0.05, and metric="euclidean". The leaf method was chosen over eom to produce more fine-grained topic clusters. The epsilon parameter merges near-duplicate clusters. Noise points (label = -1) were reassigned to their nearest cluster centroid using cosine similarity, but only if the similarity score exceeded a threshold of 0.5. Points below the threshold remained labeled as noise.

c-TF-IDF Labeling Configuration

TF-IDF was applied at the cluster level by concatenating all post texts within each cluster into a single document. The vectorizer was configured with ngram_range=(1,2), max_df=0.8, min_df=10, and max_features=10000. A combined stop word list was applied, consisting of sklearn's English stop words, custom casual/filler words, and Reddit-specific junk terms. The top 10 terms per cluster were extracted as keywords, with the top 3 forming the human-readable label. Any cluster whose top keywords matched the junk keyword list (e.g., "bot", "removed", "preview") was automatically flagged and removed.

V. Results

The pipeline successfully discovered distinct topic clusters from the r/Cambodia subreddit. After preprocessing, the usable English corpus consisted of several thousand posts. HDBSCAN identified multiple clusters of varying sizes, with the largest clusters capturing high-volume discussion themes and smaller clusters capturing niche topics. Junk clusters dominated by bot messages, moderator announcements, and image preview URLs were automatically detected and removed using the junk keyword filter.

The model was evaluated using three quantitative metrics as described in Table 2. The 2D UMAP scatter plot visually confirms that the clusters are well-separated in the projected space, with noise points shown in gray and each topic cluster in a distinct color.

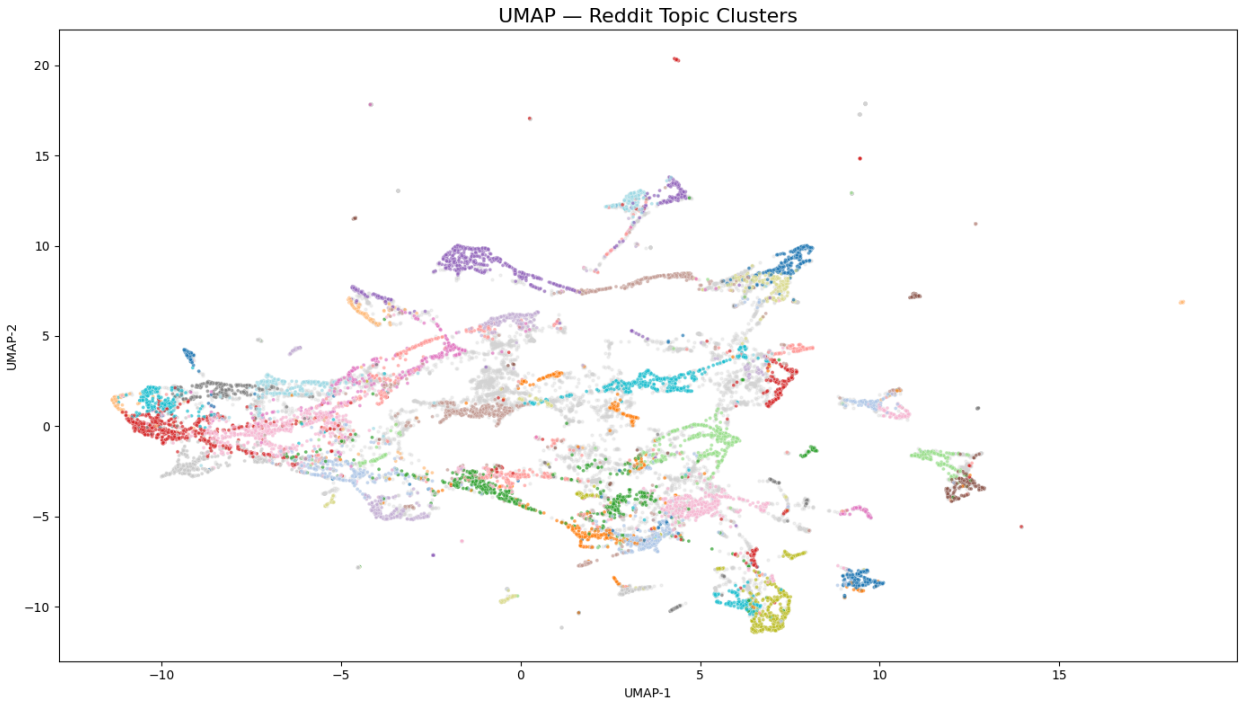
Table 2. Evaluation Metrics Used

Metric	Description
Silhouette Score	Measures how well-separated the clusters are. Range: -1 to 1, higher is better.
Davies-Bouldin Score	Measures average cluster compactness vs. separation. Lower is better.
Calinski-Harabasz Score	Ratio of between-cluster to within-cluster variance. Higher is better.
Noise Ratio	Proportion of points not assigned to any cluster by HDBSCAN.
Corpus Coverage	Proportion of posts successfully assigned to a meaningful cluster.

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Silhouette (no noise): 0.28543073
Silhouette (with noise): 0.0129336
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	value
n_clusters	46.0000
n_noise_original	9542.0000
noise_ratio_original	0.5683
silhouette_score	0.2854
davies_bouldin_score	1.1526
calinski_harabasz_score	2610.4478

The 2D UMAP visualization provides a qualitative view of cluster separation. Each colored group represents a discovered topic, while gray points represent noise or ambiguous posts. The visual separation of clusters confirms that the embeddings capture meaningful semantic structure in the subreddit discussions.



VI. Discussion

The pipeline demonstrates that transformer-based embeddings combined with density-based clustering can effectively discover meaningful topics from a noisy, community-driven Reddit corpus without any labeled data. The use of all-mpnet-base-v2 produced semantically rich embeddings that HDBSCAN could cluster more meaningfully compared to what bag-of-words methods such as TF-IDF + LSA would produce on their own.

One of the key challenges encountered was noise contamination in the early clustering attempts. Bot messages, moderator announcements, and low-quality short posts survived initial cleaning and formed their own junk clusters. This was addressed through a combination of stricter preprocessing (boilerplate regex patterns, minimum token length filter) and a post-clustering junk detection step using the keyword blacklist. The noise reassignment step using a cosine similarity threshold of 0.5 also helped retain borderline posts that would otherwise be discarded entirely.

The decision to filter to English-only posts was a practical trade-off. While the r/Cambodia subreddit does contain some Khmer text, the volume was small and langdetect produced frequent misdetections on short Khmer posts. Restricting to English allowed us to use a higher-quality embedding model. A limitation of this approach is that Khmer-language discussions are excluded from the analysis, which may miss culturally specific topics discussed primarily in Khmer.

Another limitation is that the dataset's timestamp column (created_utc) contained no valid data, which prevented any temporal analysis of topic trends over time. Additionally, the evaluation relies primarily on internal clustering metrics (Silhouette, Davies-Bouldin, Calinski-Harabasz), which measure geometric cluster quality but do not directly assess whether the discovered topics are

semantically meaningful to a human reader. Topic coherence scores or human evaluation would strengthen the assessment.

For future work, we suggest exploring BERTopic as a comparison model, which integrates transformer embeddings with a more principled topic representation approach. Additionally, incorporating Khmer language support through a multilingual model such as LaBSE, combined with better Khmer tokenization, would allow analysis of the full corpus. Temporal trend analysis could be pursued if a dataset with valid timestamps becomes available.

VII. Conclusion

This project developed a fully unsupervised topic discovery pipeline for the r/Cambodia Reddit community. The pipeline combined sentence-transformer embeddings (all-mpnet-base-v2), UMAP dimensionality reduction, HDBSCAN clustering, and c-TF-IDF cluster labeling to extract interpretable topic keywords from real-world social media data without any labeled training data.

The results show that the pipeline can successfully identify distinct, meaningful topic clusters from noisy Reddit discussions, with automatic detection and removal of junk clusters. The approach demonstrates the practical value of modern NLP techniques for community analysis and trend discovery in localized online spaces such as r/Cambodia.

The key contribution of this work is a complete, reproducible unsupervised NLP pipeline tailored to short, noisy social media text, with built-in noise handling and junk cluster detection. The findings could be useful for content moderation, community management, and understanding what topics matter most to the Cambodian online community.

VIII. References

[1] S. Stieglitz, M. Mirbabaie, B. Ross, and C. Neuberger, "Social media analytics – Challenges in topic discovery, data collection, and data preparation," International Journal of Information Management, vol. 39, pp. 156–168, 2018.

[2] D. M. Blei, A. Y. Ng, and M. I. Jordan, "Latent Dirichlet allocation," Journal of Machine Learning Research, vol. 3, pp. 993–1022, 2003.

[3] S. Deerwester, S. T. Dumais, G. W. Furnas, T. K. Landauer, and R. Harshman, "Indexing by latent semantic analysis," Journal of the American Society for Information Science, vol. 41, no. 6, pp. 391–407, 1990.

[4] T. K. Landauer, P. W. Foltz, and D. Laham, "An introduction to latent semantic analysis," Discourse Processes, vol. 25, no. 2–3, pp. 259–284, 1998.

[5] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in Proc. NAACL-HLT, 2019.

[6] L. McInnes, J. Healy, and S. Astels, "hdbscan: Hierarchical density based clustering," Journal of Open Source Software, vol. 2, no. 11, p. 205, 2017.

[7] O. Em, P. Makmee, and P. Wongupparaj, "Marketing activities among social media users in Cambodia: Mixed method research," FWU Journal of Social Sciences, vol. 17, no. 1, pp. 46–61, Spring 2023.

[8] GeeksforGeeks, "Latent Semantic Analysis (LSA)," Jul. 26, 2025. [Online]. Available: https://www.geeksforgeeks.org/machine-learning/latent-semantic-analysis/

[9] GeeksforGeeks, "How Machine Learning is Used on Social Media Platforms in 2024," [Online]. Available: https://www.geeksforgeeks.org/blogs/how-machine-learning-is-used-on-social-media-platforms-in-2024/